Supporting transitions for interacting in Digital Twins.

PhD offer at the University of Toulouse.

Supervisors:

- E. Dubois, M. Serrano, University of Toulouse, IRIT-Elipse
- C. Fleury, IMT Atlantique, INUIT
- L.-P. Bergé, CEA Tech Occitanie

Context

Digital twins are virtual representations of real-world products, systems, or processes, enabling simulation, integration, testing, monitoring, training and maintenance. They play a pivotal role in optimizing complex systems across a wide range of domains, from industrial manufacturing and energy to environmental monitoring and healthcare.

Interacting with a Digital Twin (DT) of a structure—such as a building—typically involves four entities: the user; a large, heterogeneous set of data; a physical referent (PR), which may refer to the actual structure or a physical model; and a digital replica (DR), such as a two-dimensional map or a three-dimensional model.

Situated visualization has been shown to enhance the understanding and analysis of data. Within the context of DTs, interaction with situated visualization may occur in close proximity to either the physical or digital representation of the structure. However, managing multiple workspaces concurrently can be cumbersome and cognitively demanding, potentially negatively impacting user efficiency.

Thesis Objectives

The objective of this PhD is to investigate how transitions between the physical referent (PR) and the digital replica (DR) of a Digital Twin can be supported to enhance interaction with Digital Twins (DTs).

This work will concentrate on the design and evaluation of interaction techniques that:

1. Facilitate and assist the migration of data layouts between the physical referent (PR) and digital replica (DR) of a Digital Twin;
2. Enable access to and utilization of the workspace—defined during interaction with the DR (e.g., window arrangements, data-referent links, input modalities)—in the proximity of the PR, and vice versa;
3. Address and leverage visual occlusions caused by physical objects or digital clutter to improve user experience and data accessibility.

The contributions of this research will be extensively evaluated through experimental studies applied to a range of use cases. Collaborating partners are actively engaged in several Digital Twin projects, including supervising a production hall, controlling an energy management system of a building or a plant and monitoring geological evolutions.

These partners also have extensive experience with physical models representing districts or buildings and with the use of "Fischertechnik" devices. Such devices offer a promising platform for developing robust and compelling prototypes, thereby supporting the practical validation of proposed interaction techniques.

Work Environment

The PhD candidate will be co-supervised by Emmanuel Dubois and Marcos Serrano - University of Toulouse, Cédric Fleury - IMT Brest and Louis-Pierre Bergé - CeaTech Occitanie.

The PhD student will be hosted in Toulouse within the Elipse group (https://www.irit.fr/ELIPSE/site/) at the IRIT lab. Recents works are illustrated with video on YouTube (https://www.youtube.com/@elipseIRIT/videos).

Qualifications

Required

- Master's degree in HCI, VR, Computer Science or other relevant fields
- Experience with AR / VR technologies (software, hardware)
- Experience in HCI design and prototyping
- Knowledge in 3D graphics,
- Programming skills: c#, Unity
- Written and oral communication skills in English and French
- Analytical and problem-solving abilities

Preferred

- Experience with game engines (Unity, Unreal Engine) or 3D frameworks
- Knowledge of user experience design and evaluation methods
- Previous research experience in HCI or VR/AR/XR or related fields

Application Process

Your application should include:

1. **Cover Letter**: Explaining your motivation and research interests
2. **Curriculum Vitae**: Including academic background and relevant experience
3. **Academic Transcripts**: Master's degree and relevant coursework
4. **Research Statement**: Brief description of your research interests and goals (1-2 pages)

5. **References**: Contact information for two academic or professional references

Please send your application by email at Emmanuel.Dubois@irit.fr

More information

This PhD is part of the Engineering Digital Twin program (EDT program), funded by the France 2030 investment plan. EDT is a national initiative aimed at advancing the foundations of digital twin engineering in France and Europe. By bringing together leading academic and industrial partners, EDT seeks to strengthen the bases for the design, use, and deployment of digital twins, addressing key open challenges in model hybridization, composability, development methodologies, digital coupling, and human–twin interaction.

The PhD candidate will benefit from a stimulating scientific and industrial environment of the highest level, with access to a national network of leading research institutions and industry partners, regular interactions with the broader EDT community through workshops, seminars, and joint demonstrators, and the opportunity to contribute to Artemis, the program's open software platform.

This PhD offers the opportunity to:

- Develop highly sought-after skills in system modeling, real-time data processing, and collaborative innovation.
- Collaborate with leading partners (Inria, CEA, CNRS, etc.) and validate your research on real-world industrial use cases.
- Join a network of PhD candidates within the EDT program, fostering collaboration, peer support, and interdisciplinary exchanges.
- Contribute to an open-source platform (Artemis) and publish in international conferences and journals.
- Gain recognition in a rapidly growing field, with career prospects in academic research, industrial R&D, or entrepreneurship.

Upon completion, you will be positioned as a recognized expert in a key domain for industry and research, with diverse professional opportunities in France and internationally.

References

- Mohamed Reda Benkhelifa, Emmanuel Dubois, Louis-Pierre Bergé, Marcos Serrano, Augmented urban models: Investigating how data layout and link to the physical referent impact visual search in augmented reality, *International Journal of Human-Computer Studies* (**IJHCS 2025**), Volume 203, 2025, 103556, ISSN 1071-5819, Redirecting
- Gary Perelman, Marcos Serrano, Emmanuel Dubois. Exploiting Physical Referent Features as Input for Multidimensional Data Selection in Augmented

Reality. *ACM Transactions on Computer-Human Interaction*, Volume 31 Issue 4 (ACM **ToCHI 2024**). (https://doi.org/10.1145/3648613)